Project 1 Routing for Agent KIO

Presentation Monday 26 October 2026 | Teams of 3 | Maximum 8 minutes

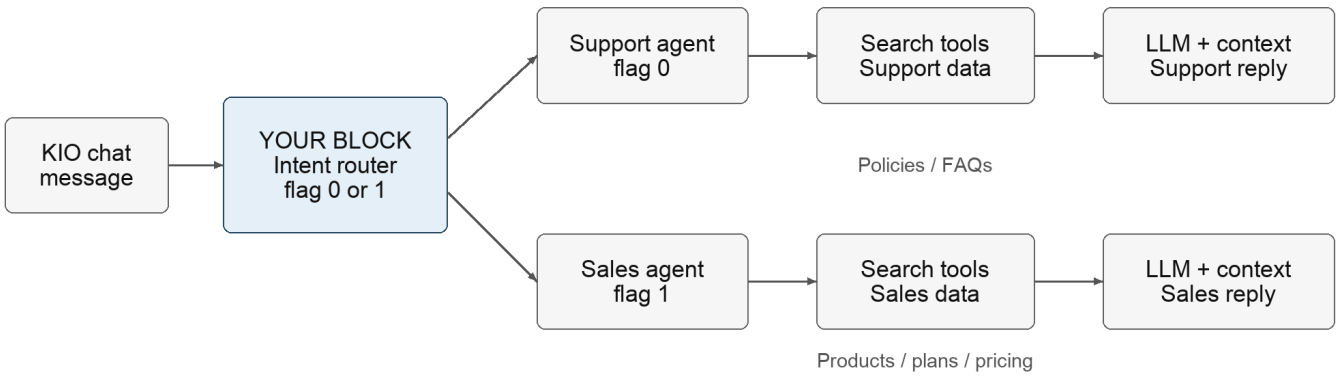
Business scenario: you work at an agency using Agent KIO, a customer-facing chatbot built on an LLM and search tools. Customers ask about existing-service problems and new purchases. The agency currently routes requests using a trained ML classifier. Your task is to evaluate whether a decision-model alternative could improve routing so customers receive relevant help and sales inquiries reach the right team.

The multi agent system and your role in RAG

KIO is the chat entry point to a system with two specialist agents. The support agent handles problems, complaints and existing requests; the sales agent handles new purchases, service packages and quotes. Each agent can use an LLM and relevant tools. When a tool retrieves knowledge and supplies it to the LLM to generate an answer, that workflow uses retrieval augmented generation (RAG).

Your block is the intent router: message in, routing flag out. Use 0 = support and 1 = sales. KIO uses the flag to call the selected agent. Routing selects the workflow; retrieval finds useful information within it. A wrong route can select the wrong tools or knowledge base.

Agent KIO architecture with two specialist agents



Build the highlighted router and simulate both agents with functions that print their names. The LLM, tools and retrieval blocks show the wider application; a full chatbot or RAG pipeline is outside this first project.

Two approaches to the same routing decision

Traditional ML: train one classifier on numeric message features. Choose a linear classifier, logistic regression with degree-2 polynomial features, or a support vector machine (SVM) with a linear or polynomial kernel. Explain the model and settings.

Decision service: compare your classifier with a model that returns customer_service or sales from the message and routing instructions. Jev from TypeSafe AI is one Decision API example, not a required provider. You may use an open-source classification or decision model with a structured-output wrapper. Identify its license, setup and output validation; it need not share Jev's architecture. Map both approaches to the same flag.

Create a concise dataset and run the study

Generate 60 synthetic English messages: 30 support and 30 sales. Include paraphrases, negation and mixed intent; prioritize existing problems in mixed requests. Agree on labels before testing. Split into 40 training and 20 test messages, balanced by class, keeping near-duplicates together. Show your generation method, labels, split and 4–6 sample rows in the presentation.

After preparing the dataset , make a comparative analysis between the two approaches .

Three students and three presentation chapters

Presenter	What to explain and show
Student 1	Problem statement and dataset: explain the routing problem, dataset generation, labeling and split. Show sample messages and explain conceptually how Jev or a similar decision model differs from your trained ML classifier.
Student 2	Jev or an alternative: explain the chosen model, inputs and instructions. Show outputs and test results; discuss successful routes, errors and what the outcomes demonstrate.
Student 3	Traditional ML classifier: explain the model, features and training. Show its boundary, outputs and test results; discuss errors, compare the two approaches and recommend one for KIO.

Choose your own time split and slide count. All three students must speak. Rehearse with transitions and demonstrations. The presentation stops at 8 minutes; no extra time is allowed.

Presentation requirements

Present all work, results and sources in your presentation. No separate files are required. Use saved results and rehearse to stay within 8 minutes.